HOW DOES INCREASING BACKPACK WEIGHT AFFECT THE WALKING POSTURE OF UNIVERSITY STUDENTS?

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1. BACKGROUND

Backpacks are a staple of student life. Filled often with textbooks, laptops, and personal items, they place a significant strain on the body over time.

At USYD, students frequently spend extended hours seated in study spaces, with **poor posture** while also commuting across campus with **heavy loads**. This combination of prolonged sitting and weight-bearing is linked to musculoskeletal discomfort, fatigue, and reduced concentration—factors that gradually impact both well-being and academic performance.

Heavy loads and poor posture are strongly associated with chronic back and neck pain, spinal issues. By examining these effects, this research aims to develop practical recommendations for **safe backpack use** and **optimal weight limits**, which ultimately supports long-term health.

This study targets **USYD students** specifically, as they represent a diverse and active university population where the impact of backpack weight and study posture can be meaningfully assessed in relation to academic demands and lifestyle habits.

Previous studies have only focused on studying school aged children and the impact of carrying backpack 15% their weight (Nagymáté et al.,2018). Not much study exists investigating university students carrying heavier bags over long distances (Chansirinukor et al., 2001). This study is especially relevant as more university students return to in-campus learning post COVID.

2. METHODS

Sampling Method

1. Stood near the Carslaw entrance and approached people walking by
2. Asked if they were university students and whether they would be willing to participate in our experiment without revealing too much detail (reduce bias)
3. Confirm that the selected participant regularly carries backpacks and do not have any back conditions
4. Inform participants of their right to withdrawal at any time and reassure them that their data will remain confidential
5. Explain the walking route, how long to walk for from the entrance of Carslaw to the Fisher Library cafe

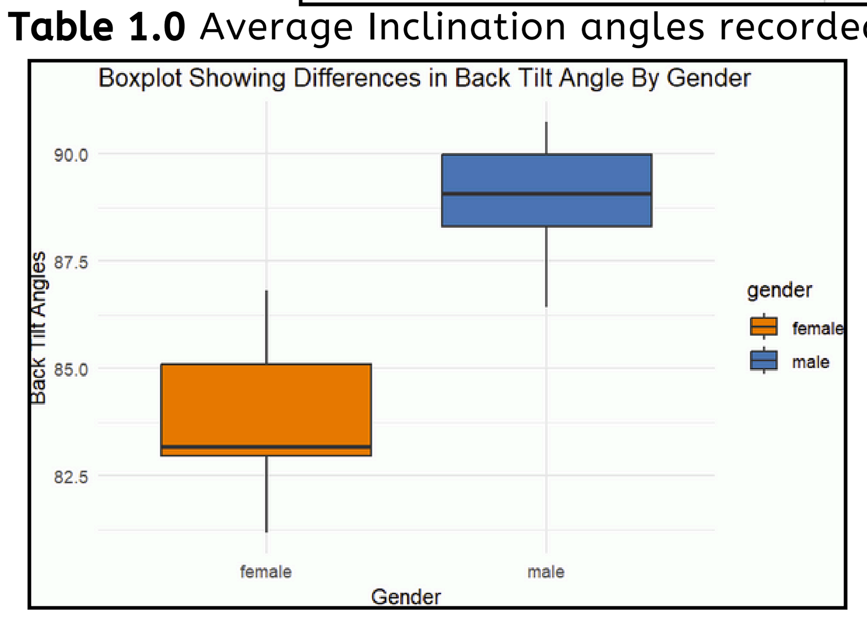
Experimental Setup:

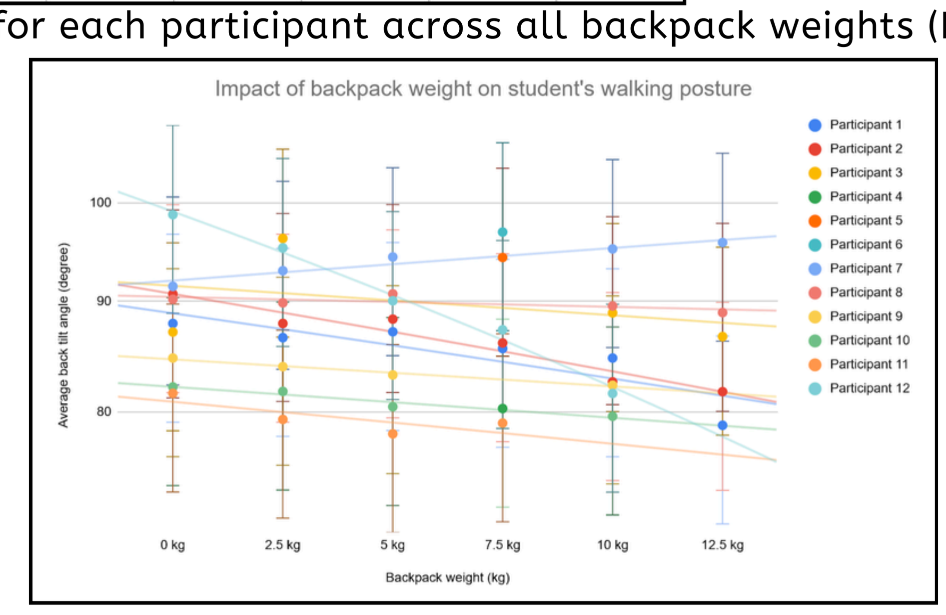
1. Downloaded the Phyphox app and select "Inclination" function on a smartphone
2. Secure the phone on the participant's upper back using a belt
3. Provide the participant with a backpack with a specific weight (2.5 kg, 5 kg, 7,5 kg, 10 kg, and 12.5 kg), using books and a laptop to achieve the target weight
4. Record the inclination data as the participants walks the assigned route
5. Walk with the participant during the trial to ensure participants' safety
6. Repeat the process twice for each participant

3. RESULT / FIGURES

Participant \ Backpack weight	0	2.5	5	7.5	10	12.5
Participant 1	87.9079	86.5859	87.1329	85.5874	84.7252	78.8583
Participant 2	90.7225	87.9089	88.318	86.097	82.6027	81.7393
Participant 3	87.1036	96.2645			88.9148	86.6976
Participant 4				80.2839		
Participant 5				94.3268		
Participant 6				96.9278		
Participant 7	91.4736	93.0124	94.3789		95.2013	95.8491
Participant 8	90.2183	89.8742	90.7395		89.5682	88.9324
Participant 9	84.7312	83.9581	83.2106		82.3047	
Participant 10	82.1542	81.7639	80.4283		79.6187	
Participant 11	81.594	79.3501	78.1383	79.0405		
Participant 12	98.7512	95.3099	90.0634	87.3208	81.5793	

Table 1.0 Average Inclination angles recorded for each participant across all backpack weights (Kg)



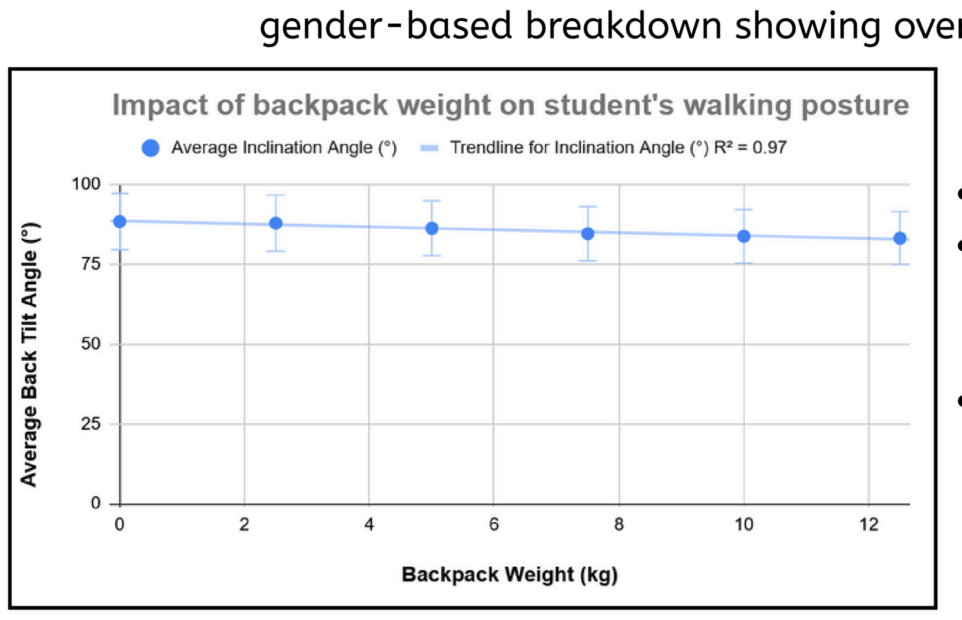


Graph 1.0 & 1.1 Trend-lines and relationship between increased backpack weights (Kg) and back tilt angle for all the participants in this study and a boxplot separating differences in male and female participants

Backpack Weight (kg)	Avg Inclination (°)	Male Avg (°)	Female Avg (°)
0.0	88.3	89.49	86.81
2.5	87.8	90.73	85.1
5.0	86.2	90.14	82.96
7.5	84.5	88.64	83.18
10.0	83.7	88.2	81.17
12.5	83.1	86.42	80.92

Backpack Weight (kg)	Average Inclination Angle (°)
0	88.3
2.5	87.8
5	86.2
7.5	84.5
10	83.7
12.5	83.1

Table 2.0 & 2.1 Average inclination angles across increasing backpack weights (0–12.5 kg), along with a gender-based breakdown showing overall averages across all participants.



Graph 2.0 Average back tilt angle vs increased backpack weight (Kg)

Observed Trends:

- Average Tilt Angle is around 83 - 88 degrees
- Tilt Angle was higher in males than females meaning they were capable of carrying heavier weights than females (better posture)
- Optimal weight is around 2.5 to 7.5 kg given the average tilt angle stays almost constant with less fluctuation on graph 1.1 & 2.0

4. DISCUSSION

The results revealed a **strong negative correlation (r = -0.985, R² = 0.97)** between backpack weight and inclination angle, confirming that posture consistently worsens as weight increases. The most pronounced change occurred **between 2.5 kg and 7.5 kg**, indicating a key threshold where postural compensation becomes significant.

correlation coefficient	-0.9850

Standard Deviation	
Male	1.5459
Female	2.1641

Males maintained a more upright posture with an average inclination angle of approximately **89.5° (SD = 1.55)**, while **females** exhibited lower angles averaging around **83.15° (SD = 2.16)** and showed greater variability. This indicates that female students may be more susceptible to posture-related strain and imbalances, aligning with *Nagymáté et al. (2018)*, who found that poor posture in females is often linked to increased risk of balance issues and musculoskeletal stress.

This study's overall findings align with *Chansirinukor et al. (2001)*, who observed similar forward-leaning posture shifts in school children as backpack weight increased — particularly in the sagittal shoulder angle, which relates directly to our focus on upper back inclination.

Sources of error included:

- Subtle belt slippage or off-axis phone placement due to body type differences
- Anatomical movement (e.g., breathing, soft tissue) affecting phone angle
- Participants unintentionally leaning before/after walking when unaware recording had started

To mitigate this:

- The first and last 5 seconds of data were trimmed
- 5 trials per participant were recorded
- The phone was repositioned for better stability mid-study

Despite these limitations, the **consistency of trends** across participants reinforces the **validity of our findings**. These results underscore the need for **posture-conscious practices** and support setting safer backpack weight limits, especially for students carrying loads daily.

Entrance of Carslaw

5. CONCLUSION

Female participants showed **greater variation in tilt angle**, hinting at possible **sex-based differences** in postural response, which may warrant further investigation. The tilt observed in this study reflects real-life academic loads and student movement patterns, making the results highly relevant for campus health and wellbeing strategies. Given the high correlation and observed trends, this study underscores the importance of load awareness and proper backpack ergonomics. **Future recommendations** for students may include limiting backpack weight, adjusting both shoulder straps correctly, and integrating posture-friendly practices throughout the academic day to reduce long-term health risks of back pain, shoulder and spinal injury.

This study demonstrated a strong negative correlation **(r = -0.985)** between **backpack weight** and **upper back inclination** angle among USYD students. As the **backpack load increased** from **2.5 kg to 12.5 kg**, participants exhibited **progressively greater forward tilt,** indicating a measurable change in posture.

These findings suggest that heavier backpacks **significantly impact spinal alignment** during walking, even over short, flat distances.

6. REFERENCES

- Chansirinukor, W. *et al* (2001) 'Effects of backpacks on students: Measurement of cervical and shoulder posture', *Australian Journal of Physiotherapy*, 47(2), pp. 110-116. doi:10.1016/s0004-9514(14)60302-0.
- Nagymáté, G., Takács, M. and Kiss, R. M. (2018) 'Does bad posture affect the standing balance?', *Cogent Medicine*, 5(1). doi: 10.1080/2331205X.2018.1503778.